

QCID: A Quantum Consciousness Interface to Probe Higher-Dimensional Physics

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September 3, 2026

Abstract

We propose the QCID Principle: Quantum Consciousness Interface to Higher Dimensions. It states that while matter with rest mass is confined to a 3-dimensional brane, massless information patterns such as consciousness can be interfaced to a Quantum Computer and made to experience simulated higher-dimensional physics. As a first proof-of-concept, we derive and simulate Maxwell's equations in 4 spatial dimensions and show that radiation intensity falls as $1/r^3$. We then outline a pathway to implement this on quantum hardware and connect it to future Brain-Computer Interfaces. This provides a testable first step toward experiencing higher-D reality.

1 Introduction

The Standard Model is formulated in 3+1 dimensions. This choice is often taken as a given, but it deserves scrutiny. If consciousness can be represented as an information pattern, and if that pattern can be hosted on a quantum substrate, then the substrate is not bound to the same spatial constraints as biological matter.

We therefore ask: what would the laws of electromagnetism look like if there were one additional spatial dimension? Photons are massless. By the QCID Principle, massless fields are the natural candidates to probe dimensions beyond our own.

2 Electrodynamics in Four Spatial Dimensions

We begin with Maxwell's equations in vacuum. In 3 spatial dimensions they lead to the wave equation:

$$\nabla_3^2 \mathbf{E} - \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0 \quad (1)$$

To extend this to 4 spatial dimensions, we add a fourth orthogonal coordinate, w . The Laplacian generalizes as:

$$\nabla_4^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + \frac{\partial^2}{\partial w^2} \quad (2)$$

Thus the 4D wave equation becomes:

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + \frac{\partial^2}{\partial w^2} - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) \mathbf{E} = 0 \quad (3)$$

For a point source, energy must spread over the surface of a 3-sphere. Therefore the intensity of radiation falls off as:

$$I_{4D}(r) \propto \frac{1}{r^3} \quad (4)$$

In contrast, in our universe $I_{3D}(r) \propto 1/r^2$.

Figure 1: Comparison of light intensity falloff in 3D vs 4D. 3D follows $1/r^2$, 4D follows $1/r^3$.

3 Quantum Simulation of 4D Physics

We simulate Eq. 3 on a quantum computer using a 4D grid. Each grid point holds a qubit representing the electric field. Time evolution is performed via Trotterization. Current NISQ devices can simulate up to 16^4 grids, which is sufficient for a QCID proof-of-concept.

The next step is BCI integration: mapping human neural patterns to the 4D simulation so the user can interact with the 4D world.

4 Discussion and The Path to BCI

This work provides a falsifiable prediction. If photons were constrained to propagate in 4D, they would obey Eq. 4. Photonic metamaterials are already used to simulate higher-D physics.

The pathway to QCID is in 3 stages: 1. Simulation, 2. Encoding via BCI, 3. Embedding of consciousness on QC. Limitations include the absence of gravity and current BCI bandwidth. Future work will address 4D gravity, where $F \propto 1/r^3$ and stable orbits do not exist.

5 Conclusion

The QCID Principle proposes that while mass is confined to 3D, information is not. By using a quantum computer as an interface, a consciousness may experience higher-dimensional physics. We provided the first mathematical and computational foundation for this idea.

References

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